Hawaii's Fuel Cost Problem: What the LSFO–LNG Price Comparison Really Shows

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Key Findings

- **A 2024 contract renegotiation has already saved Hawaiian Electric customers tens of millions — and its value will grow with the current oil price spike.** The prior supply agreement amplified crude oil price swings, adding costs beyond what global markets implied. The renegotiated terms have saved an estimated **$70–75 million** since mid-2024; if oil prices remain elevated, they will shield customers from roughly **$45–55 million per month** — about **8–10 cents per kWh** — compared to COVID-era terms set in 2022 would have cost. Customer bills will still rise roughly **8 cents per kWh** as storage tanks refill with costlier oil; under the old contract, that increase would have been closer to **18–20 cents**.
- **LNG is cheaper than oil — but the margin is narrower than it first appeared, for two reasons.** Hawai‘i has no pipeline; LNG must be liquefied, shipped across the Pacific, and converted back to gas on arrival, bringing delivered costs to roughly **$14–18 per million BTU** at current oil prices versus Hawaiian Electric's oil at **$22–23**. But the gap is also narrower than it looked before 2024: a significant part of what made LNG attractive was the old contract's unfavorable terms, not a durable gap between oil and gas costs. The renegotiation has attenuated that rationale.
- **The resulting fuel savings are modest and shrink as renewables grow.** Net of financing a $0.5 billion import terminal, savings amount to roughly **2 cents per kilowatt-hour** under slow renewable deployment, or **less than half a cent** if Hawaiian Electric's preferred buildout proceeds on schedule (see Table 1). Savings decline steeply as solar and wind displace fossil generation. Terminal costs are uncertain — large infrastructure projects in island settings commonly exceed initial estimates by 40–70% — and the broader investment case involves power plant costs beyond the scope of this fuel-cost analysis.
- **New renewables are already competitive with fossil fuel generation costs.** At current elevated oil prices, HECO's fuel cost alone runs roughly **25 cents per kWh** burning oil in its existing steam turbines. LNG fuel burned in a new, highly efficient combined-cycle gas turbine comes to about **12 cents per kWh** — but this comparison bundles a fuel switch with a generator efficiency upgrade. The fuel price difference alone, holding generator technology constant, is roughly **4–5 cents per kWh**. All of these are well above estimated all-in costs for new solar with battery storage in Hawai‘i (**9–11 cents per kWh**). At long-run futures-implied oil prices the comparison tightens further, but solar and storage remain competitive and carry zero fuel price risk.
- **Long-term LNG contracts carry risks.** Standard agreements require buyers to pay for **85–95% of contracted volumes** even if the fuel isn't burned — meaning faster renewable progress becomes a financial liability rather than a benefit. Pacific LNG prices have spiked to **$30–40 per million BTU** during supply crises, and sellers routinely seek renegotiation precisely when buyers most need relief. As a small buyer, Hawai‘i would face such negotiations with little leverage. The potential gain — at most **2 cents per kWh** in favorable scenarios — is modest and front-loaded; the downside risks are larger in magnitude and arrive under the worst conditions.

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1. Introduction

Hawai‘i’s electricity rates are the highest in the nation — roughly three times the U.S. average — and have been for decades. The main reason is straightforward: the state’s electric utilities burn imported oil to generate most of their power, and it costs a great deal to ship petroleum to remote islands in the middle of the Pacific.

Hawaiian Electric generates most of O‘ahu’s electricity by burning low-sulfur fuel oil (LSFO) at its Kahe and Waiau plants, purchased from Par Pacific’s Kapolei refinery under a long-term supply contract. A proposal from JERA Co., Japan’s largest power producer, would replace this arrangement with liquefied natural gas (LNG): a new gas-fired power plant fed by an offshore floating import terminal. The total investment is estimated at approximately $1.5–2 billion. JERA has cited roughly **$0.5 billion** for the dedicated import terminal (floating storage and regasification unit, marine berth, and pipeline connections), with an estimated **$1.0–1.5 billion** for new gas-fired generating capacity, and commercial operation targeted for 2030.

The central economic argument for LNG is that it is cheaper to burn than fuel oil. That argument has merit, but it requires two qualifications that this brief examines in detail. First, LNG’s delivered cost to Hawai‘i — including liquefaction, transoceanic shipping, and conversion back to gas on arrival — is far higher than the mainland U.S. gas prices often cited; the honest comparison is all-in LNG against what Hawaiian Electric actually pays for oil. Second, and less widely recognized: part of what made Hawaiian Electric’s fuel oil look so expensive in recent years was not the global oil market but the terms of its own supply contract. Those terms were unusually unfavorable — the contract’s steep pricing structure amplified crude oil price swings rather than dampening them. A 2024 renegotiation substantially improved those terms, narrowing the cost gap with LNG and delivering tens of millions of dollars in savings before a single drop of LNG was imported. Understanding both of these qualifications is essential to evaluating the LNG proposal on its actual merits.

This analysis is limited to fuel pricing. It emerged from work to refine energy cost inputs for a comprehensive grid optimization study using the Switch model — and the details of HECO’s fuel supply contract, particularly the 2024 renegotiation, proved consequential enough to warrant a standalone treatment. Whether LNG is the right investment when weighed against the full portfolio of alternatives — accelerated solar and wind development, energy storage, demand response, and grid flexibility — requires that broader analysis, which we are pursuing separately.

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2. How HECO’s Fuel Bill Works

Hawaiian Electric purchases LSFO under a contract with Par Pacific that ties the price to a global marine bunker fuel grade called very low-sulfur fuel oil (VLSFO), which trades actively at the Singapore and Houston bunker markets. VLSFO itself is tightly linked to global crude oil prices, so the mechanical chain runs: *global crude markets → VLSFO spot price → HECO contract price*.

The contract does not simply pass through the daily VLSFO price. The contract sets HECO’s purchase price based on a rolling average of recent VLSFO prices — so a sudden market spike works its way into HECO’s fuel cost only gradually, across many deliveries. On top of that, HECO stores fuel in large tanks and draws on that inventory over time, so what customers pay in any given month reflects the blended cost of fuel purchased over many preceding weeks — not just the most recent delivery. These two layers of smoothing together explain why a rapid price surge like that of early 2026 appears in customer bills only gradually, as older, cheaper inventory is depleted and replaced at higher prices. This buffering works in both directions: it also delays the benefit when prices fall. The transition between contract regimes adds a further wrinkle: HECO appears to have drawn down much of its inventory before the renegotiation, allowing it to rebuild under the new terms — which is why the full effect of the renegotiated contract took several months to show up in reported prices.

The contract's pricing formula has two key parameters: the *slope* (how much HECO's price rises for each dollar increase in VLSFO) and the *intercept* (a base level that provides a floor for Par Pacific's revenue). These parameters have changed substantially over the contract's history.

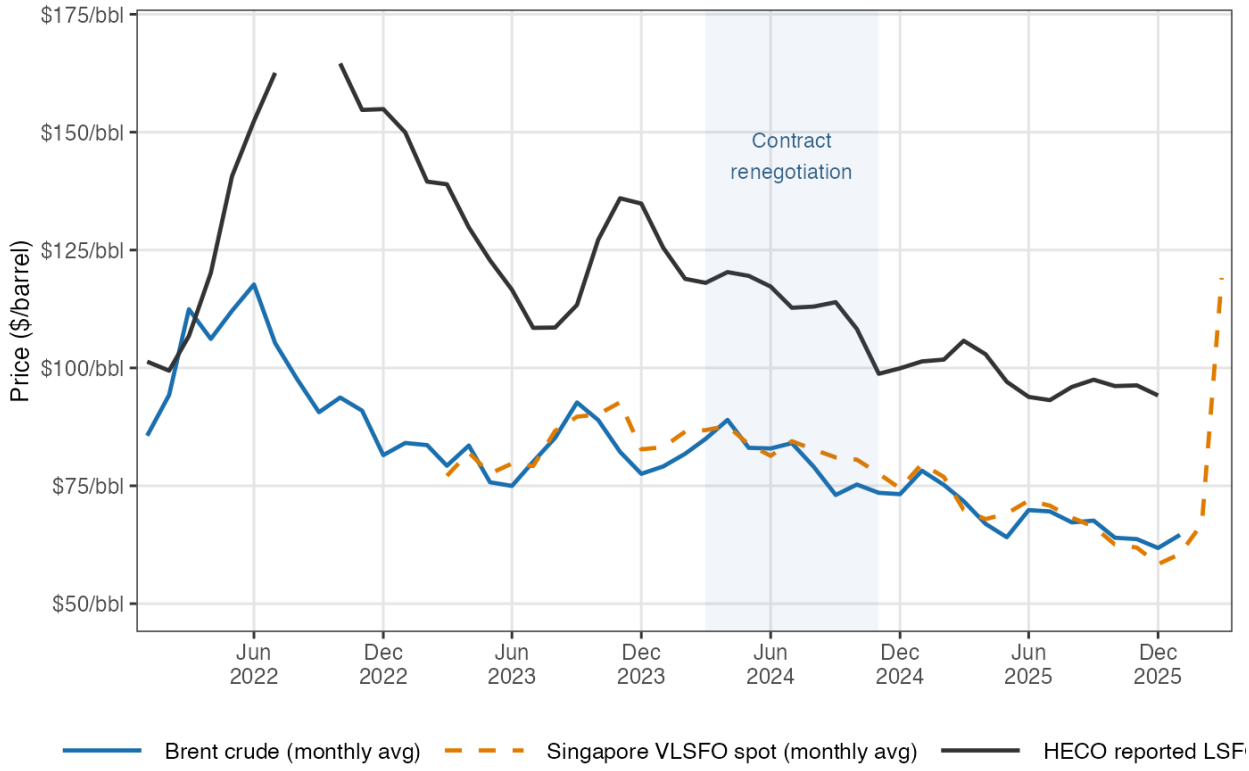

Figure 1. Monthly Brent crude, Singapore VLSFO spot price, and HECO reported LSFO purchase price, 2022–2026. All series in $/barrel. The shaded band marks the contract renegotiation window (March–November 2024). Sources: FRED, Ship & Bunker, EIA Form 923.

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3. The Par Contract: A Complex History

COVID, IMO 2020, and an Unusual Starting Point

In January 2020, new International Maritime Organization (IMO) regulations took effect requiring oceangoing ships to switch from high-sulfur fuel oil to low-sulfur alternatives. This abruptly elevated VLSFO prices relative to crude oil, as refineries scrambled to produce more low-sulfur product. Then, two months later, the COVID-19 pandemic collapsed demand for aviation fuel, gasoline, and marine fuel simultaneously. Par Pacific, HECO's refinery supplier, faced severe financial stress as demand for nearly all of its products evaporated at once.

The fuel supply contract renegotiated in this environment — which we refer to as the R1 regime — gave Par Pacific a generous margin structure: a steep slope (approximately 1.9 dollars of LSFO price increase for each dollar increase in the VLSFO benchmark) and a large negative intercept. In plain terms, Par received roughly $1.90 in additional LSFO revenue for every dollar that the VLSFO benchmark rose. The commercial logic was sound: Par needed assurance that the refinery would remain economically viable. But it meant that HECO customers were bearing substantial price risk on the upside.

From 2022 through early 2024, as global energy prices surged following the Russian invasion of Ukraine and global supply chains tightened, this contract structure amplified the pass-through to HECO customers. LSFO prices reflected not just the commodity cost of crude oil but also Par's elevated margin entitlement.

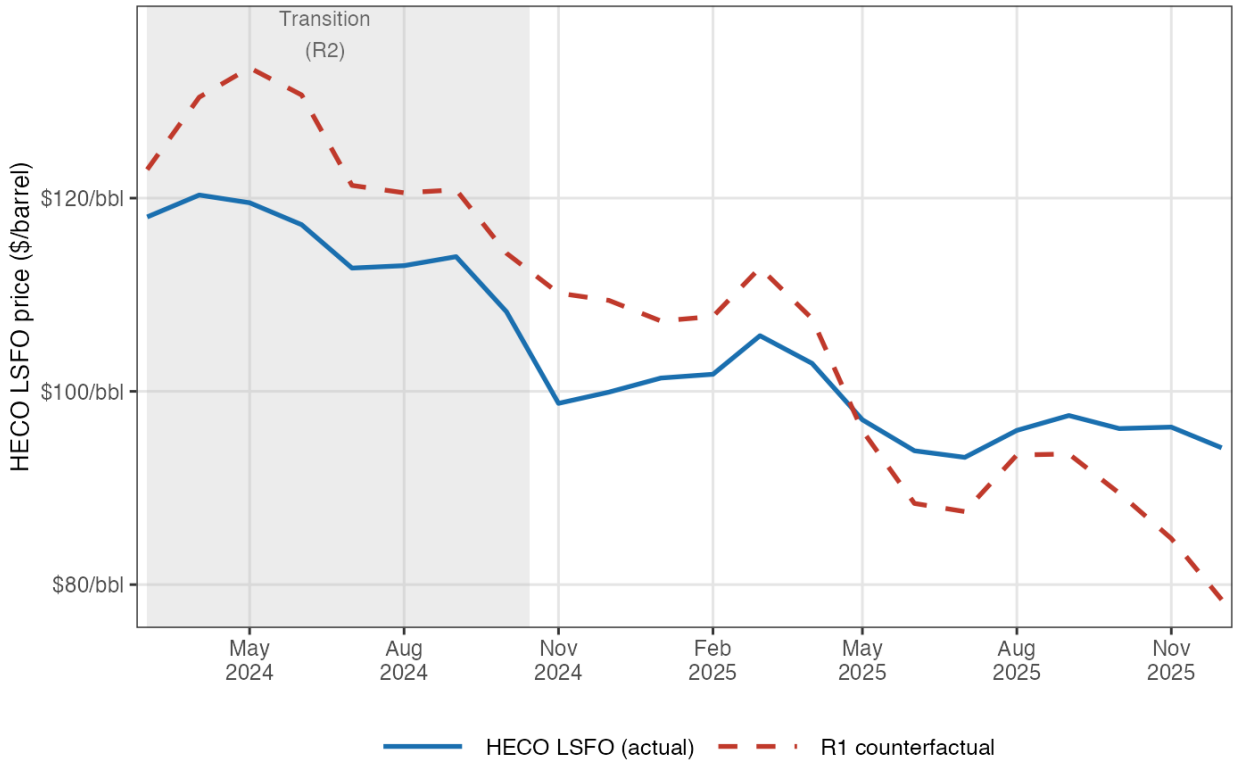

Figure 2. HECO reported LSFO price (blue) vs. estimated R1 counterfactual price (red dashed), March 2024–December 2025. The vertical gap between the lines is the monthly saving (or cost) from the renegotiation. The grey band marks the R2 transition period. Sources: EIA Form 923, Hawaiian Electric filings, UHERO model.

The 2024 Renegotiation

The contract was revised in 2024 — what we term the R3 regime. While exact terms are confidential, their effect is clearly visible in public LSFO price data. The new structure has a substantially flatter slope (~0.74, compared to ~1.90 under R1) and a positive intercept (~$37/barrel). In practice, this gives Par Pacific a better price floor when crude is low — providing some protection against prolonged price slumps — while capping the LSFO price premium that HECO customers pay when crude is high. It is a more symmetric risk-sharing arrangement than R1.

To quantify the difference, we estimate what HECO's LSFO costs would have been under R1 terms during the post-renegotiation period and compare it to actual R3 costs. The cumulative difference through the end of 2025 amounts to approximately **$70–75 million in savings** for HECO customers (measured against R1 terms negotiated under acute COVID-era financial stress at Par Pacific; whether those terms would have persisted in a normalized market is uncertain).

More dramatically, the current VLSFO price spike — driven by refining disruptions associated with the Iran War — illustrates precisely the kind of exposure the R3 renegotiation was designed to limit. The R3 and R1 formulas cross at a VLSFO price of roughly **$83/barrel**: below that level, R1 would actually be cheaper; above it, R3's savings grow proportionally with the market price. At $145–150/barrel VLSFO (the mid-March 2026 level), the implied monthly fuel cost difference between R3 and R1 amounts to roughly **$45–55 million**

per month — approximately **8–10 cents per kWh** across total electricity sales. Because customer bills reflect the average cost of fuel in HECO's storage tanks — not the price of the most recent delivery — this has not yet passed through to bills; the tanks still contain fuel purchased before the spike. If VLSFO remains elevated, customers will see roughly **8 cents/kWh** added to their bills relative to January 2026 — about a 20% increase in a typical bill — as the averaging window catches up. Under R1 terms, the equivalent fuel cost increase would have been roughly **18–20 cents/kWh**, or about 45–50% of a typical bill. These figures are based on Brent-implied VLSFO projections; if the current VLSFO-Brent spread persists, actual costs and savings will be larger still.

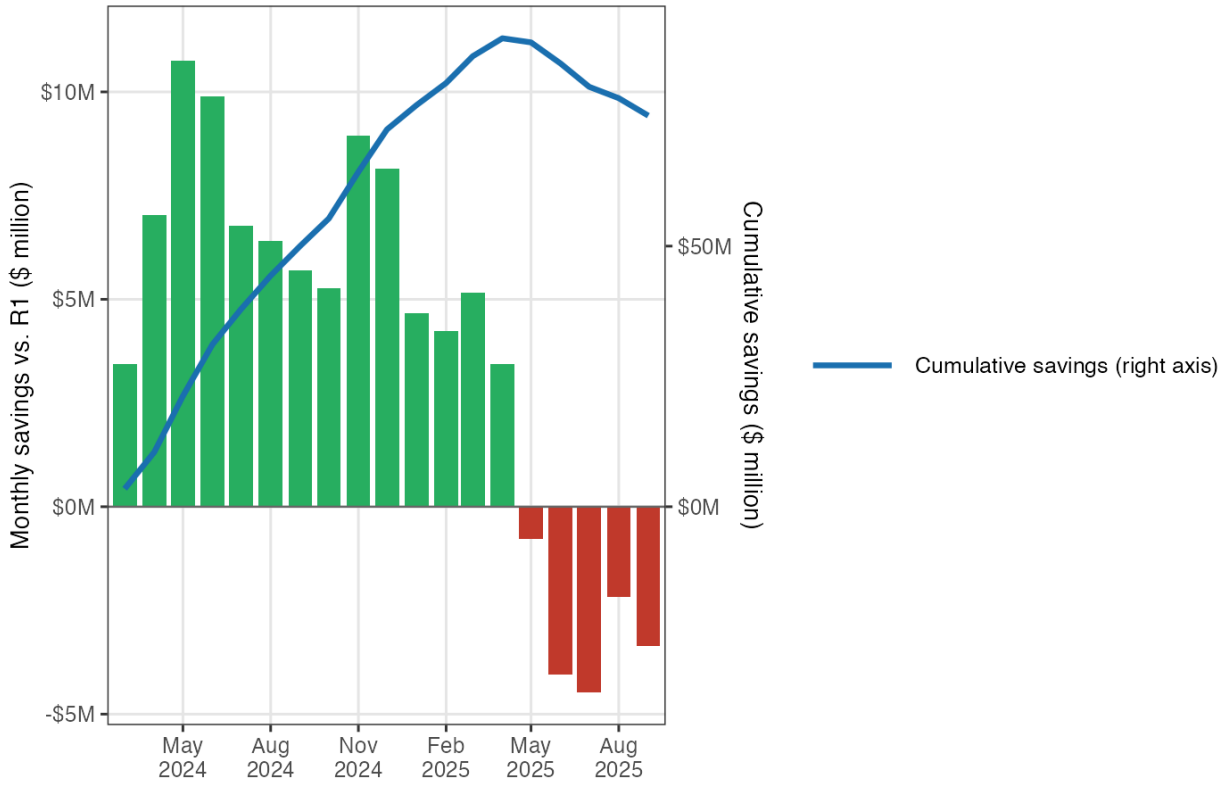

Figure 3. Monthly fuel cost savings (green) or additional costs (red) under the R3 contract relative to the R1 counterfactual, and cumulative total (blue line, right axis). Negative months reflect periods when crude prices were low enough that R1's steeper slope would have produced lower LSFO prices. Sources: EIA Form 923, Hawaiian Electric filings, UHERO model.

This history has a direct implication for any LNG contract negotiation: the structure of the pricing formula — its slope, its floor, its averaging window — has been worth tens of millions of dollars annually to HECO customers, with the benefit growing sharply as VLSFO rises above the ~$83/barrel level where the R3 and R1 formulas cross. Getting the contract structure right matters at least as much as the headline price comparison.

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4. The LNG Price Comparison: What the Numbers Show

The Basic Formula

Hawai'i cannot access natural gas through a pipeline. LNG must be liquefied at an export terminal, loaded onto specialized tankers, shipped to an import facility, and regasified before it can be burned. Each step

adds cost. The HSEO feasibility work, conducted by Facts Global Energy (FGE) in August 2024, identified **0.4 million tons per year (mtpa)** as roughly the minimum throughput at which a floating storage and regasification unit (FSRU) could be economically viable — large enough to spread fixed infrastructure costs, while not necessarily requiring customers beyond HECO. At that scale, regasification adds approximately **$3.93 per MMBtu** on top of the commodity cost (assuming near-capacity throughput; fixed FSRU costs — vessel lease, crew, maintenance — are largely independent of volume, so effective per-MMBtu regas costs would be substantially higher at the ~25% average utilization implied by the preferred IGP scenario). To put 0.4 mtpa in perspective: burned in a new combined-cycle gas turbine (6.5 MMBtu/MWh), it could supply roughly **3,200 GWh per year** — more than half of HECO's current annual net electricity sales of approximately 6,100 GWh. Burned in a simple-cycle combustion turbine (about 9.5 MMBtu/MWh), the same LNG volume would yield roughly **2,200 GWh per year**, or about a third of net sales. Whether 0.4 mtpa matches actual dispatch requirements depends on the generation mix, the pace of renewable buildout, and the role of other thermal resources — questions addressed in Section 6.

HSEO's August 2024 feasibility work cited an indicative contract price for LNG delivered to Hawai'i of approximately **0.118 times the Brent crude price plus $0.60 per MMBtu**, with regasification added separately. As of March 31, 2026, Brent futures were trading at approximately **$113 per barrel** (front-month contract), translating to an all-in delivered cost of approximately **$17.9/MMBtu** for a 0.4 mtpa facility.

HECO's steady-state LSFO cost at the same Brent price, once its inventory averaging window fully rebuilds under the current R3 contract structure, is approximately **$22.2/MMBtu**. The fuel cost difference favoring LNG is thus approximately **$4.3/MMBtu** at current prices.

JERA has also indicated that it may structure the contract with pricing indexed to Henry Hub, the U.S. natural gas benchmark, rather than to Brent crude. Under a standard Henry Hub-linked LNG structure — roughly 115% of Henry Hub plus liquefaction, shipping, and regasification costs totaling around $8/MMBtu — the all-in delivered price at current Henry Hub prices (~$3.50/MMBtu) would be approximately **$12/MMBtu**, a wider advantage against LSFO. Henry Hub is not, however, a stable benchmark: it exceeded **$7/MMBtu in January 2026**, which would raise the all-in Hawai'i cost to roughly **$16.50/MMBtu** — comparable to LSFO at moderate oil prices. The formula price advantage depends on where the index sits, not just its average level.

The Long-Run Picture

The economic advantage of LNG over LSFO depends heavily on oil prices. At low crude prices, LSFO becomes relatively cheap, and the case for LNG narrows. Our long-run model, which traces HECO's steady-state LSFO cost as a function of the Brent price under the current R3 contract, shows that LNG **fuel** (Brent-indexed, HECO-only facility scale) is cheaper than LSFO at any Brent price above roughly **$70–75 per barrel** — a threshold comfortably below even the EIA's downside price scenarios for the 2030s.

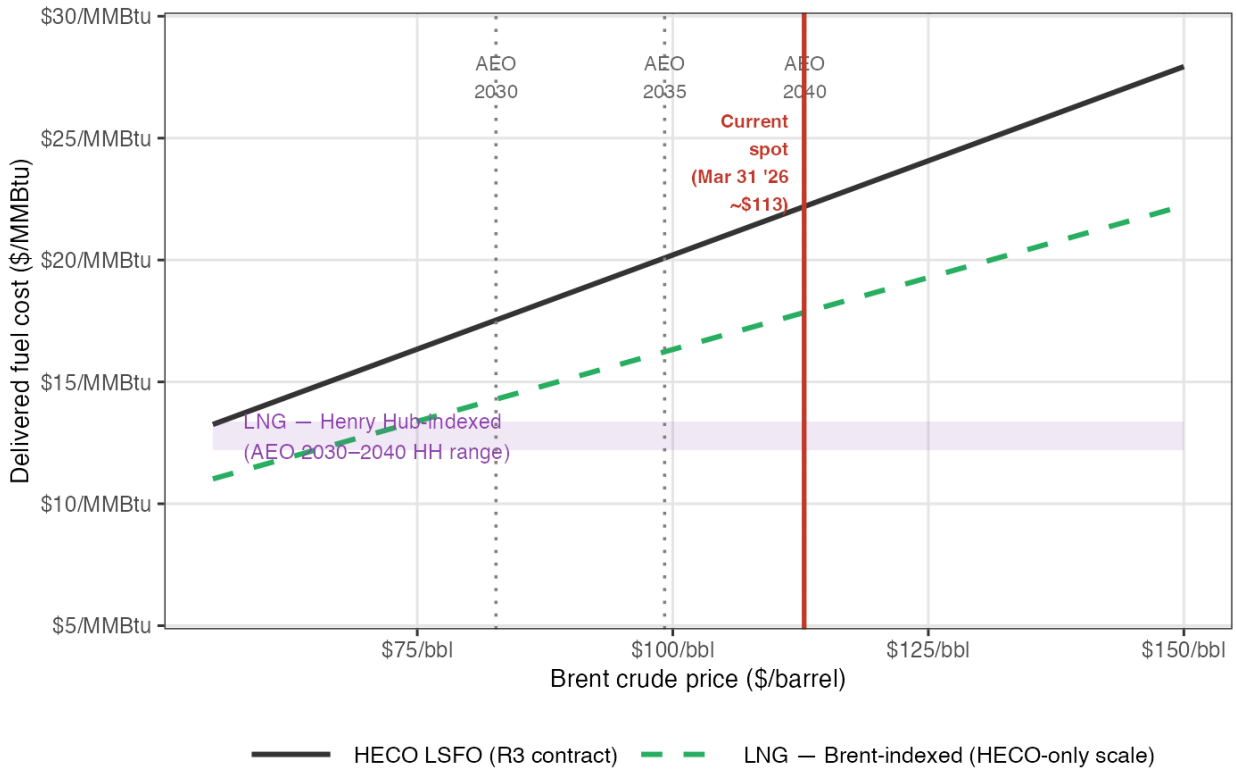

Figure 4. Long-run steady-state HECO LSFO cost (R3 contract, black) vs. LNG all-in delivered cost — Brent-indexed (green dashed) and Henry Hub-indexed range (purple band, spanning AEO 2030–2040 Henry Hub). The red vertical marks current Brent (~$113/barrel, March 31, 2026), at which LNG saves approximately $4/MMBtu relative to steady-state LSFO. Grey dotted verticals show the EIA AEO 2026 reference path for 2030 ($83), 2035 ($99), and 2040 ($113). Note that Brent futures as of March 31 imply the 2030 price (~$72/bbl) will be well below the AEO reference. All costs in $/MMBtu. Sources: UHERO model, HSEO/FGE August 2024 indicative formula, EIA AEO 2026, Investing.com.

The Forward Path

In the near term, HECO's LSFO costs are rising as its inventory rebuilds under the R3 contract structure. The R3 contract uses a longer lookback window on VLSFO prices than R1 did; as HECO replenishes its inventory to the new, larger target level, the averaging window lengthens and the reported purchase price rises — even if the underlying VLSFO price stays flat. Figure 5 shows actual observed costs for January and February 2026, the model's transition path for March, and a forward projection for April–June based on Brent futures prices as of March 31. On this basis, HECO's LSFO cost rises from approximately $89/barrel in January to around $115/barrel by June, with the inventory rebuild accounting for much of the increase above what Brent prices alone would imply.

This projection should be read as a central estimate with material uncertainty in both directions. The underlying Brent-to-VLSFO relationship, estimated statistically, has a root mean squared error of roughly $4/barrel at daily frequency; more importantly, its residuals are strongly autocorrelated — when VLSFO departs from its Brent-implied level, the gap tends to persist for weeks or months rather than reverting quickly. As of mid-March 2026, VLSFO was running roughly **$25–35/barrel above** what Brent prices alone would predict, likely driven by the same supply disruption that has pushed crude higher. If VLSFO remains elevated relative to Brent, HECO's actual costs will exceed the projected path shown in Figure 5 — potentially reaching $140–146/barrel at full averaging-window equilibrium, equivalent to roughly $22–23/MMBtu. That scenario

would widen the fuel cost advantage of LNG relative to LSFO. It would also substantially amplify the savings from the R3 contract renegotiation: the R3 contract's flatter slope (~0.74 versus R1's ~1.90) means its cost advantage over R1 grows proportionally with VLSFO, reaching roughly **$70–75/barrel** at current market prices — far larger than the historical average over 2024–2025 that underlies the cumulative savings figures cited above.

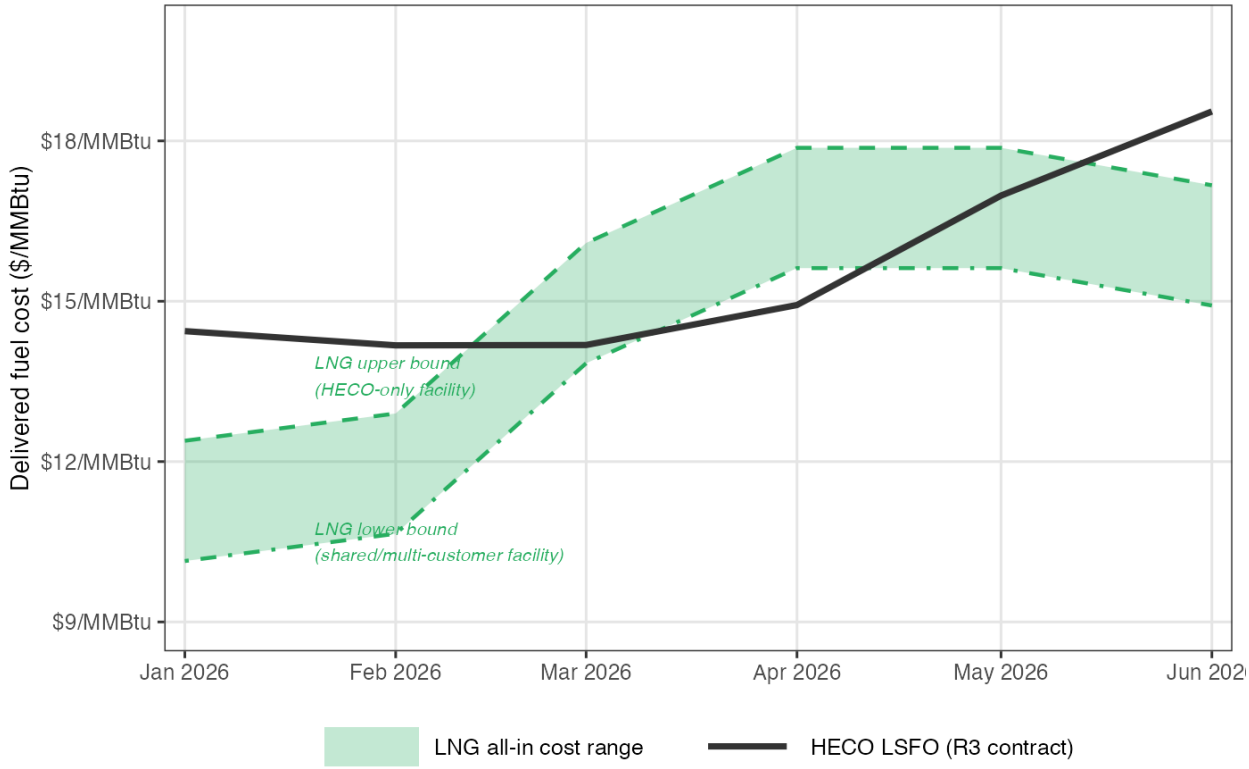

Figure 5. Projected HECO LSFO cost (R3 contract, solid line) vs. the LNG all-in delivered cost range, January–June 2026. Actual observed costs for January and February; model projection for March onward. The shaded band spans the upper bound (HECO-only facility) to the lower bound (shared or multi-customer facility); dashed lines mark each boundary. LNG prices reflect month-specific Brent: realized averages for January ($67/bbl) and February ($71/bbl), realized for March ($98/bbl), and Investing.com futures for April–June ($113, $113, $107/bbl). The LSFO projection assumes VLSFO tracks Brent according to the model relationship; if VLSFO remains elevated above Brent-implied levels (as it was in mid-March 2026), actual LSFO costs would be higher than shown. Sources: UHERO model, HSEO/FGE August 2024 indicative formula, FRED (DCOILBRENTEU), Investing.com.

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5. Henry Hub Is Not a Hawai‘i Price

Public discussion of LNG for Hawai‘i has sometimes compared LNG contract costs against continental piped gas — Henry Hub or Canadian spot prices — which have traded at $2–4/MMBtu for much of the past decade, dramatically cheaper than LSFO by a factor of three or four. But Hawai‘i has no pipeline connection to the U.S. gas network or any continental source. There is no way to receive piped gas at any price; gas must be liquefied, shipped, and regasified before it can be burned.

Delivering natural gas to Hawai‘i requires liquefying it (paying a liquefaction fee of roughly $2.75/MMBtu), loading it onto an LNG tanker (shipping costs of roughly $1.50/MMBtu on Pacific routes), and regasifying it

on arrival. The economically meaningful price for Hawaiʻi is the Japan-Korea Marker (JKM), the Pacific LNG benchmark, or equivalently the all-in cost of a long-term delivered LNG contract. Over the past decade, JKM has averaged $8–12/MMBtu, and has spiked far higher.

The reason for extreme spikes is structural. Liquefaction facilities — the plants that convert pipeline gas into LNG for export — are among the most capital-intensive pieces of energy infrastructure in the world, and new capacity typically takes five to ten years to permit and build. Global liquefaction capacity is therefore relatively fixed over any horizon of a few years. When demand surges — from an unusually cold winter, a geopolitical disruption, or an outage at a major export terminal — there is no quick way to add supply. Cargoes are instead diverted to the highest-paying buyer, and spot prices adjust until demand falls back to meet the constrained supply. This is an inherent feature of the LNG market, not an anomaly.

The relevant comparison is between **LNG all-in** and **HECO's LSFO cost**. On that basis, as we showed above, LNG is indeed cheaper at current prices — but the advantage is $4–6/MMBtu rather than $14–18/MMBtu. This distinction matters enormously for assessing whether the investment pencils out.

The Indexation Question

If a Hawaiʻi LNG contract were indexed to Henry Hub rather than Brent, would this insulate customers from LNG price volatility? Partly, but not fully.

First, Henry Hub is not itself stable. It traded between $1.60 and $9.00/MMBtu over the past decade, with sharp winter spikes during cold snaps. In **January 2026**, Henry Hub exceeded **$7/MMBtu**, which would have produced an all-in delivered cost to Hawaiʻi of roughly **$16.50/MMBtu** — comparable to LSFO costs under moderate oil prices and far above the benchmark figure of ~$12/MMBtu that circulates in discussions of HH-indexed LNG contracts. At $9/MMBtu Henry Hub, the all-in cost reaches approximately **$18.50/MMBtu**, higher than LSFO in most scenarios. The formula price alone is not predictably cheaper.

Second, and more fundamentally, the formula price is not necessarily the price Hawaiʻi would pay during a supply crisis. The Pacific LNG market shows this clearly.

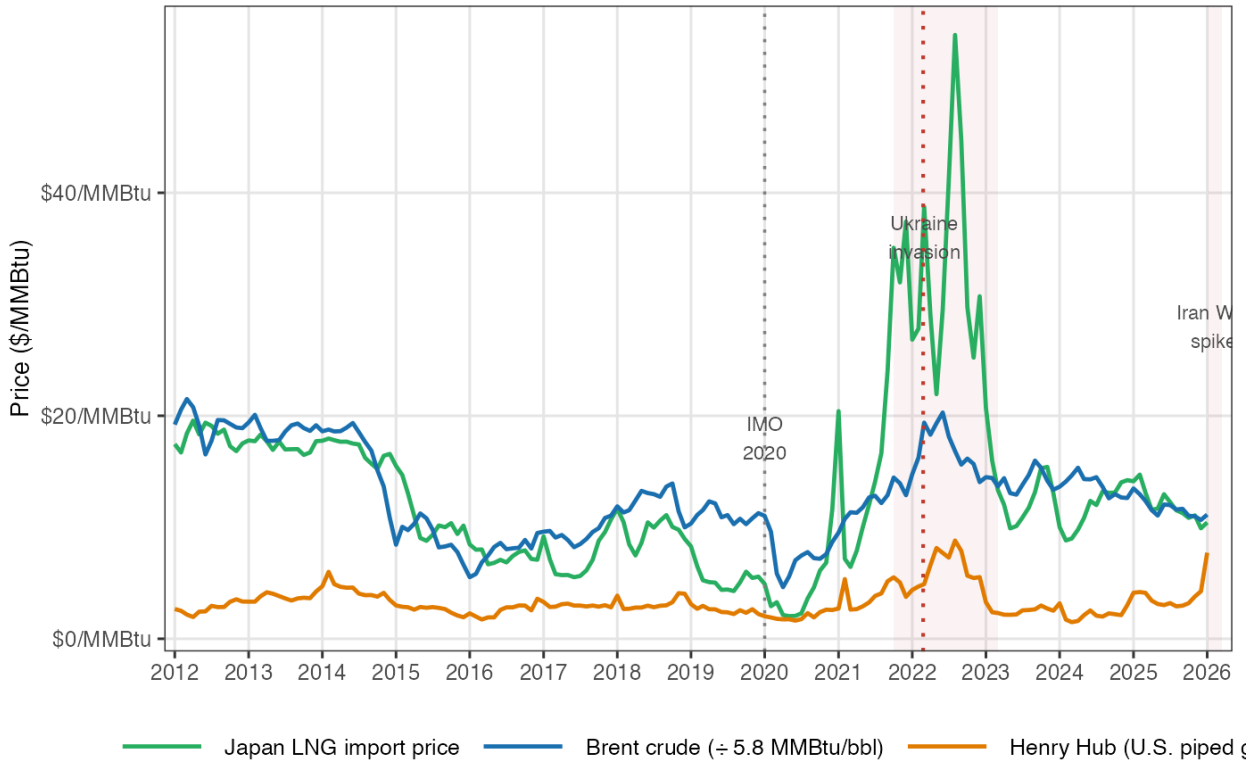

Figure 6. Japan LNG import price, Henry Hub spot price, and Brent crude (converted at 5.8 MMBtu/bbl), monthly, 2012–2026, all in $/MMBtu. During the 2022 European energy crisis, the Japan LNG price exceeded $30/MMBtu while Henry Hub remained below $10/MMBtu — a spread of more than $20/MMBtu. The Japan LNG series is a monthly average of contract and spot deliveries and understates spot JKM spikes; actual Pacific spot prices peaked above $40/MMBtu in 2022. Sources: FRED (MHHNGSP, PNGASJPUSDM, POILBREUSDM).

During the 2022 European energy crisis, the Japan LNG import price — a close proxy for Pacific spot LNG — rose above $30/MMBtu while Henry Hub was at $6/MMBtu. The spread between the two benchmarks, which is normally $5–10/MMBtu reflecting the liquefaction and shipping costs, exceeded $25/MMBtu. This happened precisely because global liquefaction capacity was fixed: Russia had curtailed pipeline gas flows to Europe, European buyers were competing aggressively for any available LNG cargo, and no new liquefaction capacity could be brought online quickly enough to relieve the squeeze. The cargoes that would otherwise have gone to Asia were redirected west, Asia paid more, and spot prices decoupled sharply from formula-price contracts. During the current Middle East supply disruption, similar dynamics are at work. Formula-price buyers are left exposed to force majeure claims or commercial pressure for renegotiation.

Renegotiation Is the Norm

Long-term LNG contracts have formal pricing mechanisms, but they are regularly revised. An analysis of Japanese, South Korean, Taiwanese, and Spanish LNG import price series finds that contract renegotiations appear as statistically identifiable structural breaks in the pricing relationship — they are not rare exceptions but a recurring feature of LNG markets (Agerton, *Energy Journal*, 2017). These renegotiations tend to occur when there is a large gap between the formula price and the prevailing spot price — precisely the conditions under which the contract's price protection would be most valuable.

The 2020 COVID episode illustrates the process: several Chinese LNG buyers issued force majeure notices, arguing that plant shutdowns prevented them from taking contracted volumes. Sellers pushed back, but the episode opened commercial conversations about contract revisions (Oxford Institute for Energy Studies, 2020). During the 2022 energy crisis, LNG exporters in tight markets faced different pressure: buyers seeking relief from formula prices that had risen along with oil, even as spot cargoes became unavailable at any price.

For Hawai'i, these dynamics create asymmetric exposure. As a small buyer — representing a fraction of a percent of global LNG trade — the state has essentially no leverage in any renegotiation. When the market turns against formula buyers, sellers have both the commercial incentive and the contractual tools (force majeure clauses, price review mechanisms) to seek better terms.

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6. What Do the Numbers Say Over the Life of the Contract?

A note on current market conditions: as of March 31, 2026, Brent crude front-month futures are at approximately **$113/barrel** — nearly double the EIA's pre-Iran War Short-Term Energy Outlook reference of $58/barrel for 2026. At this price, the steady-state LNG advantage is approximately **$4/MMBtu**. However, markets interpret the spike as temporary. Brent futures fall steeply over the next seven months — from $113 (May 2026) to $84 (December 2026) — and settle into a long-run backwardation of around **$69–73/barrel** from 2029 onward (Investing.com, March 31, 2026). This futures-implied long-run price is roughly 12–14% below the EIA AEO reference path, which projects $83/bbl in 2030 rising to $113/bbl by 2040. The futures curve therefore materially changes the economics of the 2029–2044 operating window.

To assess the investment case we use the generation mix projections from HECO's Integrated Grid Plan (IGP), which provides two scenarios spanning the plausible range of renewable deployment. Under the **preferred scenario** — solar, wind, and storage proceed at the pace the IGP considers achievable — total fossil

generation falls from about 2,400 GWh in 2029 to under 100 GWh by 2044. Under the **land-constrained scenario** — slower renewable deployment due to siting constraints — fossil generation falls from about 4,000 GWh to around 240 GWh over the same period. As noted above, a 0.4 mtpa facility at CCGT heat rates can supply roughly 3,200 GWh per year. The land-constrained scenario approaches that throughput in its early years but averages only about 55% of it over the full contract period; the preferred scenario averages roughly 25%. Both scenarios imply substantial underutilization — and therefore substantial take-or-pay exposure if contracts are written against 0.4 mtpa capacity. JERA's published economic case appears to assume annual generation on the order of 2.8 TWh/yr, which would require throughput well above 0.4 mtpa and exceeds the land-constrained IGP average by more than 55% and the preferred scenario average by more than 3×. How JERA reconciles this with HECO's own renewable deployment plans is not clear from publicly available materials, but the gap between JERA's assumed generation volumes and either IGP scenario is a significant source of divergence between the two economic cases.

The fuel savings in Table 1 reflect two distinct effects: the lower per-MMBtu cost of LNG compared to LSFO, and the higher thermal efficiency of a new combined-cycle gas turbine (**6.5 MMBtu/MWh**) compared to Hawai'i's existing oil-fired steam turbines (**~11 MMBtu/MWh**). These two effects compound, and — as Figure 7 and the discussion below make clear — the efficiency gain is the larger of the two. Note that if HECO would eventually replace its aging steam turbines regardless of fuel choice, the efficiency gain is not attributable to LNG and the net benefit would be smaller.

The correct way to compare fuel savings against capital cost is to convert the capital outlay to an annual repayment stream using HECO's financing cost (WACC), subtract that annual obligation from annual fuel savings to get the net annual benefit, and discount the net stream at 5% to obtain a present value. We use **$0.5 billion** as the capital base — JERA's stated cost for the import terminal — recognizing that this is likely a lower bound and that the power plant represents additional capital whose LNG-attributable share is uncertain. At 7–9% WACC, the terminal's annual financing cost is **$55–62 million per year**, and the present value of that stream at 5% over 15 years is **$0.57–0.64 billion**. Table 1 also reports the **breakeven total capex** for each scenario — the maximum total LNG-attributable capital (terminal plus any share of the power plant) at which the investment still breaks even, shown as a range across the two WACC assumptions.

Table 1 presents the results across both scenarios and two oil price paths.

Scenario	Avg. annual fuel savings	Net after terminal capex (¢/kWh)	Net PV (7% WACC)	Net PV (9% WACC)	Breakeven total capex
IGP preferred + AEO Brent	$85M/yr	0.4–0.5¢	+$0.62B	+$0.55B	$0.93–1.04B
IGP preferred + Futures Brent	$70M/yr	0.1–0.2¢	+$0.42B	+$0.35B	$0.77–0.87B
IGP land constrained + AEO Brent	$197M/yr	2.2–2.3¢	+$2.05B	+$1.98B	$2.04–2.30B
IGP land constrained + Futures Brent	$157M/yr	1.6–1.7¢	+$1.57B	+$1.50B	$1.66–1.88B

Table 1. Avg. annual fuel savings: gross fuel cost difference (LNG vs. LSFO), averaged over 2029–2044. Net after terminal capex (¢/kWh): annual fuel savings minus annual terminal financing cost ($55–62M/yr at 7–9% WACC on $0.5B), divided by total Oahu net electricity sales (~6,100 GWh/yr); range reflects 7–9% WACC. Net PV = PV of net annual benefits discounted at 5% real from 2026. Breakeven total capex: maximum total LNG-attributable capital at which Net PV = 0; range reflects 7–9% WACC. Power plant adds an estimated $1.0–1.5B beyond the terminal. AEO Brent: EIA AEO 2026 reference. Futures: Investing.com March 31, 2026 (avg. $69.6/bbl, 2029–2044). Heat rates: 11 MMBtu/MWh for LSFO (existing steam turbines), 6.5 MMBtu/MWh for LNG (new combined-cycle turbine). These mixed heat rates capture the full

benefit of adopting LNG — including both the fuel price difference and the generator efficiency upgrade — but approximately 65–75% of the savings shown reflects the efficiency gain rather than the fuel price difference alone. Fuel price savings only (holding generator technology constant at a simple-cycle combustion turbine rate) would be roughly 75–85% smaller than the totals shown.

Several observations follow.

At terminal-only capital, the fuel savings cover the financing requirement across all scenarios. Annual savings of $70–85M/yr (preferred) and $157–197M/yr (land-constrained) both exceed the $55–62M/yr financing cost of a $0.5B terminal. Even the weakest case — preferred scenario under futures oil prices — produces a net PV of +$0.35–0.42B. The breakeven column in Table 1 shows how much total LNG-attributable capital each scenario can sustain before going negative.

The power plant is the critical unknown. Whether the $1.0–1.5B power plant cost is attributable to the LNG decision depends on whether Hawaiian Electric would build new generating capacity anyway as its aging steam turbines retire. If the answer is yes — new gas-fired plant would be needed for reliability regardless of fuel — only the $0.5B terminal is LNG-specific and the economics look solid. If the power plant is contingent on LNG, the preferred + futures case reaches its breakeven at $0.77–0.87B total capex, leaving only $0.27–0.37B of headroom above the terminal cost — less than 25% of the power plant. Beyond the attribution question, the proposed plant may not be optimally sized or timed relative to what would be built in the absence of LNG; a plant dimensioned around 0.4 mtpa throughput may be larger than Hawai'i's declining fossil generation needs by the 2030s.

Cost overruns could consume thin margins in the preferred scenario. Flyvbjerg's systematic review of megaprojects finds average cost overruns of 40–70% across large infrastructure investments, with the upper tail driven by technically novel or remote projects — both of which apply here (Flyvbjerg, *Oxford Handbook of Megaproject Management*, 2014). IEA and IRENA analyses of island-state LNG terminals and FSRUs specifically note construction cost uncertainty of 30–80% above initial estimates, attributed to marine logistics, permitting complexity, and limited local contractor competition. JERA's $0.5B figure has not been independently verified or subjected to PUC review. Applying the midpoint of this range: a 60% overrun implies an actual terminal cost near **$0.8B**, which nearly exhausts the breakeven headroom in the preferred + futures scenario ($0.77–0.87B). A 70% overrun ($0.85B terminal) flips that scenario negative without any power plant attribution. The land-constrained scenarios are substantially more resilient — breakeven capex of $1.66–2.30B — meaning even a 70% overrun on the terminal leaves ample margin unless substantial power plant costs are also assigned to the LNG decision.

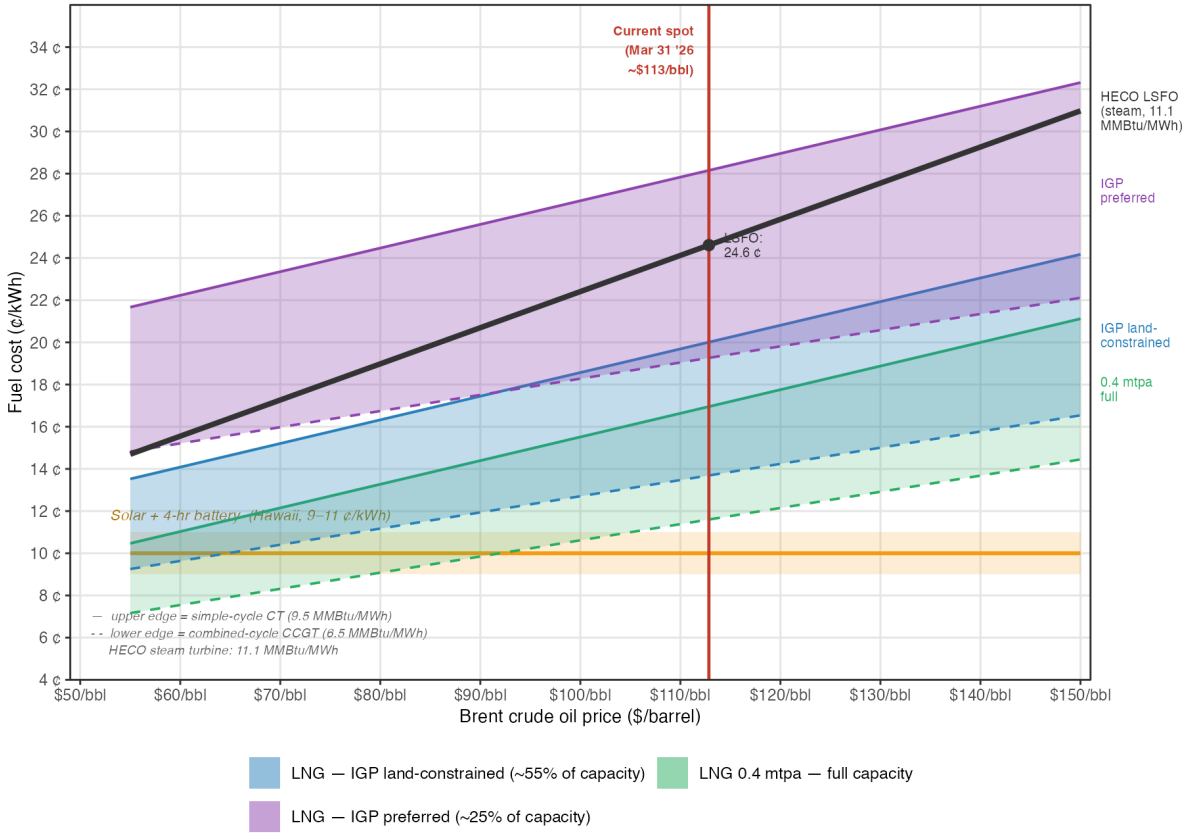
Savings decline sharply after the mid-2030s in both scenarios. Annual savings fall from roughly $230M in 2029 to under $10M by 2044 in the preferred scenario, as fossil generation approaches zero. Even in the land-constrained scenario, savings fall below the terminal's financing cost in the final years of the contract. The front-loaded savings carry the positive PV; the back half of the contract is a drag. This timing mismatch matters for HECO's financial planning and for the terms of any take-or-pay commitment.

Price-path uncertainty spans roughly $0.5B in outcomes. The AEO and futures price paths differ by about $13–40/bbl over the 2029–2044 operating window — EIA projects prices rising from $83/bbl to $113/bbl; futures markets price a flat $69–73/bbl. The resulting difference in net PV is roughly $0.4–0.5B in each scenario. Neither forecast is reliable; the futures path reflects current market consensus, which the academic literature finds more accurate on average than agency projections over long horizons.

The efficiency gain should not be fully attributed to the LNG decision. Hawaiian Electric's aging steam turbines will need replacement regardless of fuel choice. A new combined-cycle gas turbine uses about 40% less fuel per kilowatt-hour than the existing steam units (6.5 versus ~11 MMBtu/MWh), and that efficiency improvement — not the difference in fuel price — accounts for roughly **65–75% of the gross fuel savings** shown in Table 1.[1] Figure 7 makes this concrete: each colored band shows the full range of LNG fuel costs depending on generator technology, from combined-cycle (lower, dashed edge) to simple-cycle combustion

[1] At the AEO 2030 Brent reference price of $83/bbl, LSFO costs approximately 18¢/kWh at the existing steam turbine heat rate; LNG costs approximately 9¢/kWh at combined-cycle heat rates — a stated gap of ~9¢. Holding both fuels at the same simple-cycle combustion turbine heat rate (9.5 MMBtu/MWh) narrows the gap to ~2¢. Generator efficiency thus accounts for approximately 73% of the total. The efficiency share is similar across plausible Brent price paths.

turbine (upper, solid edge). The vertical width within each band represents generator technology choice — not fuel price. The black LSFO line cuts through the IGP preferred scenario band at current prices, meaning LNG saves money in that scenario *only if paired with a highly efficient combined-cycle turbine*; if a simple-cycle turbine is assumed instead, LNG fuel in the preferred scenario costs *more* than LSFO at today's prices. To the extent HECO would eventually build new generation regardless of fuel choice — and modern gas turbines can burn fuel oil at comparable efficiency — the efficiency gain is not a benefit of LNG specifically, and the net PV figures in Table 1 overstate the benefit of the fuel switch alone.



LNG: HSEO/FGE (Aug 2024) Brent-indexed formula (0.118 × Brent + $0.60/MMBtu) plus regasification. Each band spans simple-cycle CT (9.5 MMBtu/MWh, upper edge) to CCGT (6.5 MMBtu/MWh, lower edge). Regasification at full capacity: $3.93/MMBtu (0.4 mtpa), $1.68/MMBtu (1.0 mtpa). FSRU costs are largely fixed (vessel lease, crew, maintenance), so per-MMBtu cost scales inversely with utilization: IGP land-constrained (~55% avg, 308 MW = 2,698 GWh/yr) → $7.15/MMBtu; IGP preferred (~25% avg, 207 MW = 1,813 GWh/yr) → $15.72/MMBtu. LSFO: R3 contract steady-state model (UHERO), steam turbine (11.1 MMBtu/MWh). Solar+battery: NREL ATB 2030 moderate (~$70/MWh mainland) + 62% Hawaii premium (Leidos/EIA 2016) = 9–11 ¢/kWh. Current Brent: $112.85/bbl (Mar 31, 2026).

Figure 7. LNG fuel cost ranges vs. HECO LSFO vs. Hawaii solar+battery, in cents per kilowatt-hour, plotted against the Brent crude oil price. Each colored band represents one LNG infrastructure scenario; the lower (dashed) edge assumes a combined-cycle gas turbine (6.5 MMBtu/MWh) and the upper (solid) edge assumes a simple-cycle combustion turbine (9.5 MMBtu/MWh) — the vertical width of each band reflects generator technology choice, not fuel price. The black line is HECO's steady-state LSFO cost under the R3 contract (existing steam turbines, 11.1 MMBtu/MWh). The amber band is the estimated all-in levelized cost of new solar with four-hour battery storage in Hawai'i (9–11¢/kWh). Where the LSFO line falls inside a colored band, the assumed generator technology — not the fuel — determines which option is cheaper. The red vertical marks the current Brent spot price (~$113/bbl, March 31, 2026). Sources: HSEO/FGE August 2024 indicative formula; UHERO model; NREL ATB 2024; Leidos/EIA 2016.

New renewable energy is cost-competitive with fossil fuel generation — including on fuel costs alone. NREL's Annual Technology Baseline (moderate scenario, 2030) puts utility-scale solar with four-hour battery storage at roughly **$70/MWh** (≈7¢/kWh) on the mainland (NREL, 2024). A 2016 analysis by Leidos Engineering for EIA estimated Hawaii's (Honolulu's) construction cost premium for utility-scale solar PV at approximately **62%** above the national base — reflecting higher logistics, labor, and permitting costs — which puts the Hawaii-adjusted solar-plus-storage estimate in the range of **9–11¢/kWh** (EIA/Leidos Engineering, 2016). Actual project costs have varied considerably, and higher costs on some Hawaii projects may be

due in part to solvable procurement design issues. For comparison, LNG fuel cost — assuming a new combined-cycle turbine — runs roughly **12¢/kWh** at current contract-based oil prices and **8–9¢/kWh** at long-run futures-implied prices; these figures exclude power plant and terminal capital. In a simple-cycle combustion turbine, the same LNG fuel costs roughly **17¢/kWh** at current prices and **12¢/kWh** at long-run futures prices. At current elevated oil prices (~$113/bbl Brent), HECO's LSFO fuel cost alone runs approximately **25¢/kWh**, making new solar and storage clearly cheaper on a total-cost basis regardless of the generator technology assumed for LNG. At long-run futures-implied oil prices, LNG fuel in a combined-cycle turbine (~8–9¢/kWh) is broadly comparable to solar — but in a simple-cycle turbine (~12¢/kWh), LNG fuel sits at or above the top of the Hawaii solar+battery range (9–11¢/kWh). Solar carries **zero fuel price risk** and requires no long-term take-or-pay commitment. In the scenarios where LNG saves meaningfully relative to LSFO, solar saves considerably more. This comparison is central to the portfolio analysis underway and is beyond the scope of this brief.

JERA's business model may reduce the effective terminal cost — but bunkering revenue should be treated as upside, not base-case relief. JERA has cited Honolulu's position on trans-Pacific shipping lanes as a natural LNG bunkering market, and has suggested that marine bunker fuel sales and the FSRU's residual value after 2045 could partially offset terminal costs recovered from HECO ratepayers. The bunkering claim is plausible in principle: if bunkering revenue were to cover $50–100M of terminal debt service over the contract period, that represents roughly **$0.3–0.6B in present-value terms** — meaningful against a $0.5B terminal, and potentially a material buffer against cost overruns in the preferred scenario. However, LNG marine fuel adoption is contingent on sufficient fleet conversion by trans-Pacific carriers, and Honolulu's competitive position as a bunkering hub relative to other Pacific ports is unestablished. Until a shipping demand study substantiates the claim, bunkering revenue should be treated as speculative upside rather than a base-case input. FSRU residual values face similar uncertainty as the shipping industry transitions away from fossil fuels. Any cost-sharing arrangements should be fully disclosed to the PUC, with ratepayers benefiting only to the extent the allocation is transparent and contractually enforceable.

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7. Contract Risks That Could Erode the Savings

The fuel cost projections above assume that the formula price is the price Hawai'i pays throughout the contract period. Three risks could cause realized costs to diverge significantly from formula-price projections.

Supply crisis exposure. As discussed above, global LNG supply shocks have repeatedly caused Pacific spot prices to spike far above formula levels. Force majeure clauses in LNG contracts are triggered by supply disruptions at export terminals, port closures, and armed conflict — precisely the events that elevate spot prices. If a seller cannot deliver at formula terms during a crisis, Hawai'i must either accept an outage or purchase replacement cargoes on the spot market. Force majeure language in LNG contracts varies widely; recent assessments find that most contracts do not adequately cover scenarios in which quarantine restrictions, cargo diversion, or government actions — rather than physical damage — prevent delivery (Denison, *Journal of World Energy Law & Business*, 2021).

Take-or-pay and declining demand. LNG supply contracts typically require buyers to accept or pay for 85–95% of contracted annual volumes, regardless of actual consumption (Oxford Institute for Energy Studies, 2022). Hawai'i's clean energy law requires 100% renewable electricity by 2045. A 20-year LNG contract signed around 2028 would run almost exactly through the period during which thermal generation is supposed to decline to zero. If solar, wind, and storage deployment proceeds faster than projected — or if demand-side efficiency reduces load — take-or-pay obligations could require payment for LNG volumes the state does not burn. This stranded cost risk is asymmetric: faster-than-expected clean energy progress, which should be the desired outcome, becomes a financial liability under a long take-or-pay commitment.

LNG oversupply and formula lock-in. Large volumes of new U.S. Gulf Coast liquefaction capacity are coming online over 2025–2027. If global LNG supply grows faster than demand, spot prices could fall well below oil- or Henry Hub-linked formula prices, leaving Hawai'i locked into above-market terms. The LSFO

alternative, sourced domestically from Par Pacific, would not be exposed to global LNG supply dynamics in the same way.

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8. What Would a Well-Structured Contract Require?

None of these risks precludes an LNG contract, but they must be addressed explicitly in the contract terms rather than assumed away. Hawai‘i’s experience with the Par Pacific LSFO contract is instructive: the renegotiation from R1 to R3 terms delivered a flatter pricing slope and a positive intercept — effectively a price collar — that is now paying dividends in a volatile market. Similar features are achievable in an LNG contract.

Price collar: A cap on the formula price when the pricing index rises sharply, in exchange for a floor that protects the seller when the index falls. This is the most direct mechanism for limiting customer exposure in supply crises.

Narrow force majeure definitions: Seller force majeure should require evidence of physical inability to deliver, not merely that delivering at formula price would be commercially unattractive. Commercial cargo diversion — sending a cargo to a higher-paying market — should not qualify as force majeure.

Declining take-or-pay quantities: Minimum volumes should be set conservatively and should decrease on a schedule consistent with Hawai‘i’s renewable buildout trajectory under the state’s Renewable Portfolio Standard. If renewable capacity grows faster than projected, the contract should include provisions for downward quantity adjustment without penalty.

Shorter initial term: A 10-year initial contract with renewal options, rather than a 20-year commitment, would substantially reduce stranded-asset risk while still providing the long-term supply security that justifies infrastructure investment.

Public transparency and PUC oversight: Long-term fuel supply contracts for regulated utilities are ultimately a public matter. The terms of any LNG agreement should be available to the Public Utilities Commission in sufficient detail to permit rigorous assessment of the risk exposure embedded in the contract.

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9. Conclusions

The fuel price case for LNG is genuine but narrower than often presented. At current oil prices (~$113/bbl), LNG delivered to an offshore Hawai‘i terminal costs roughly **$4–5/MMBtu less** than HECO’s current LSFO contract. Whether that advantage justifies the capital investment depends on two things that cut against each other: oil prices need to stay high enough to sustain the per-MMBtu savings, and renewable energy deployment needs to be slow enough that there is substantial thermal generation left to benefit from those savings.

The two HECO Integrated Grid Plan scenarios bound the range. The terminal infrastructure — an FSRU, berth, and pipeline — has been quoted at **$0.5 billion** by the project proponent (JERA), consistent with the regasification tariffs in the HSEO/FGE analysis. At that cost basis the economics are positive in all scenarios, including the preferred renewable buildout with futures-implied oil prices (**$0.35–0.42B net PV**). The land-constrained scenario, with higher and more sustained fossil volumes, yields average annual savings of $157–197M and net PVs of **$1.50–2.05B**. But the terminal is only part of the picture. A new CCGT power plant capable of burning LNG would add an estimated **$1.0–1.5 billion** more. Whether that additional capital is properly attributed to the LNG decision — rather than to generation investment Hawai‘i would undertake regardless — is the pivotal economic question. If the full combined cost is LNG-attributable, the preferred

buildout with futures prices approaches breakeven (breakeven total capex: $0.77–0.87B), with only $0.27–0.37B of headroom for cost overruns.

The cost of capital and the discount rate play distinct roles. The WACC determines the annual repayment obligation — roughly **$55M/yr at 7% WACC** or **$62M/yr at 9% WACC**, applied to the $0.5B terminal; the present value of that capital stream ($0.57–0.64B at 5% over 15 years) is the correct benchmark against which to compare the PV of fuel savings. Attributing any portion of the power plant to the LNG investment raises both figures proportionally.

Whether a contract is indexed to Brent or Henry Hub changes the formula but not the fundamental uncertainties. Henry Hub exceeded $7/MMBtu in January 2026, producing an all-in Hawai‘i cost of approximately $16.50/MMBtu. In supply crises, Pacific spot LNG has historically diverged from formula prices by $20/MMBtu or more, regardless of indexation.

Hawai‘i's current LSFO contract demonstrates that careful contract structure can deliver real consumer protection in volatile markets. Any LNG arrangement deserves the same discipline: explicit price collars, narrow force majeure definitions, declining take-or-pay volumes aligned with the renewable buildout trajectory, and full PUC scrutiny of all material terms.

One feature of this picture deserves emphasis. Even in the scenarios that favor LNG, the net gains per kilowatt-hour are modest. Under the land-constrained buildout, where fossil volumes remain high, net annual savings net of terminal capital recovery average roughly $95–142M — translating to about **1.5–2 cents per kilowatt-hour** across Oahu's total demand (futures to AEO price path), or roughly 4–6% off a typical electricity bill. Under the preferred renewable buildout, the margin is thinner still: net annual savings average $15–30M, equivalent to **less than half a cent per kilowatt-hour**. Customers in those scenarios might barely notice the difference. The risks that cut the other way are both more numerous and larger in magnitude. A supply disruption could force Hawai‘i to replace undelivered cargoes at spot prices $15–25/MMBtu above the contracted formula — adding perhaps 10–15 cents per kWh during the crisis. Take-or-pay obligations that outlast the thermal generation they were written to serve could lock in hundreds of millions in costs even as the state meets its clean energy goals. Renegotiation risk, when it materializes, tends to arrive precisely when buyers have the least leverage. The result is an asymmetric risk profile: the upside is bounded and quiet; the downside is open-ended and arrives under the worst circumstances.

This analysis addresses only the fuel cost comparison. It does not evaluate potential grid integration benefits sometimes attributed to gas-fired generation — including the ability to accommodate solar and wind variability and reduce renewable curtailment — or the full-system costs and benefits of alternative investment paths. Those questions require grid modeling that accounts for the complete portfolio of supply and demand options, including the interactions between LNG, renewable expansion, storage, and demand response; that analysis is underway using the Switch capacity expansion model and will be the subject of a subsequent report.

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Technical Notes

LSFO price model. HECO's monthly LSFO purchase price is modeled using a nested moving-average framework calibrated to public HECO fuel cost reports (EIA Form 923) and Singapore and Houston VLSFO spot prices from Ship & Bunker. Three contract regimes (R1, R2, R3) are identified via structural break analysis, with regime-specific slopes, intercepts, inner and outer averaging windows, and an inventory-days variable tracking the rebuilding of the outer window after contract changes. The long-run steady-state LSFO price at any given Brent level is derived by setting Brent equal to WTI in the model and running the VLSFO bridge to its equilibrium.

LNG price formulas. Brent-indexed: 0.118 × Brent + $0.60/MMBtu (DES commodity, HSEO/FGE August 2024 indicative) plus $3.93/MMBtu regasification (FGE August 2024 minimum-viable-scale assumption, 0.4 mtpa FSRU). Henry Hub-indexed: 1.15 × HH + $2.75 (liquefaction) + $1.50 (shipping) + $3.93 (regasification), consistent with typical U.S. LNG export contract structures (IEA, 2025).

PV calculation. Annual fossil generation by scenario derived from HECO Integrated Grid Plan (IGP, 2023) generation mix shares — preferred and land-constrained scenarios — linearly interpolated between anchor years (2022, 2030, 2035, 2040, 2045) and applied to projected Oahu net demand (IGP 8760-hour demand file). Annual net benefits = annual fuel savings (fossil GWh × (LSFO cost/MWh − LNG cost/MWh)) minus annual capital recovery (CRF(WACC, 15yr) × $0.5B terminal). Net benefits discounted at 5% real to base year 2026. Heat rates: 11 MMBtu/MWh (existing oil steam); 6.5 MMBtu/MWh (new LNG CCGT). Brent price paths: EIA AEO 2026 reference, and Investing.com futures (March 31, 2026). Operating window: 2029–2044. Power plant costs ($1.0–1.5B) are not included in the base capital assumption; breakeven total capex is reported separately as the maximum LNG-attributable investment at which net PV equals zero.

Savings vs. R1 counterfactual. Monthly savings estimated as the difference between predicted R3 LSFO costs and predicted R1 costs at observed VLSFO prices, scaled by EIA Form 923 Oahu oil-fired generation volumes. Uncertainty reflects estimation error in the moving-average window parameters.

UHERO, the University of Hawai‘i Economic Research Organization, provides objective economic analysis and data for Hawai‘i. The views expressed are those of the authors and do not represent any state agency. Data and replication code available at [repository link].